

A. HARSHAN

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EDUCATION

B.Tech – Artificial Intelligence and Data Science

Chennai Institute of Technology, Chennai, Tamil Nadu

2024 – 2028 | Chennai, Tamil Nadu
Batch: 2024 – 2028

SKILLS

Programming

Python, C++, Java

Web Technologies

HTML, CSS

Tools & Technologies

OpenCV, TensorFlow, PyTorch, Flask, FastAPI

Concepts

Machine Learning, Deep Learning, Computer Vision

CERTIFICATIONS

- Introduction to Modern AI
- Introduction to IoT and Digital Transformation
- Introduction to Data Science
- AI Fundamentals with IBM SkillsBuild
- Introduction to Cybersecurity

INTERNSHIP EXPERIENCE

Saint-Gobain India Pvt. Ltd. (INDEC 4.0)

Duration: 21 Days

Worked on backend development and digital transformation tools. Gained hands-on experience in building scalable APIs and handling real-world data systems.

PROJECTS

Object Detection Model using Ultralytics YOLO

Developed an object detection system using YOLOv8 for real-time multi-object recognition. Integrated OpenCV for video stream processing. Applied data augmentation and optimization techniques to improve model accuracy.

Voice Assistant using Python

Built a voice-controlled assistant using SpeechRecognition, pyttsx3, and Google TTS. Automated tasks such as web search, application launching, and system operations via voice commands.

Web Backend Development – Saint-Gobain Intern Project

Developed RESTful APIs using Flask and FastAPI. Connected APIs to PostgreSQL/MySQL databases for efficient data handling. Improved backend scalability and ensured secure data transactions.

Face Recognition System using AI

Developed a face recognition system using TensorFlow. Implemented deep learning for face detection and identification. Enhanced system performance under varied lighting and angles.

Finger Detection System using AI

Built a real-time finger detection system using TensorFlow and OpenCV. Used image processing techniques like edge detection for accurate tracking. Optimized performance for dynamic environments and gesture handling.

AI-Based Subaquatic Visual Enhancement for Ocean Monitoring

Developed an AI-based system to enhance underwater image clarity and visibility. Applied techniques like dehazing, color correction, and contrast enhancement. Used OpenCV and TensorFlow/PyTorch for real-time video enhancement. Improved marine object detection accuracy in low-light and turbid conditions. Optimized system for real-time deployment in ocean monitoring applications.